

Virtual Trackball on VR Controller: Evaluation of 3D Rotation Methods in Virtual Reality

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ABSTRACT

Rotating 3D objects is an essential operation in virtual reality (VR). However, efficient rotation methods with the current VR controllers have not been considered extensively yet. Users must repeatedly move their arms and wrists to rotate an object with the current VR controller. We considered utilizing the trackpad available in most VR controllers as a virtual trackball for an efficient rotation method and implemented two types of virtual trackballs (Arcball and Two-axis Valuator) to enable additional rotation using the thumb while holding an object with a VR controller. In this study, we investigated whether a controller with a virtual trackball would be effective for 3D manipulation tasks. The results showed that participants could perform the tasks faster with Arcball but not faster with Two-axis Valuator than with the regular VR controller. Also, most participants preferred Arcball to Two-axis Valuator and felt Arcball more natural than Two-axis Valuator.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

virtual trackball, rotation method, VR controller

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1 INTRODUCTION

With the popularization of virtual reality (VR) technology, it has become more frequent to interact with 3D virtual objects in a VR environment. Most commercial VR systems provide the virtual hand technique [10] for manipulation. Users can manipulate 3D virtual objects with hand-held controllers or hand tracking system. While the virtual hand technique could be beneficial since it is intuitive and natural, it could be unfavorable when rotating an object to a large extent because the users have to grab, turn, and release the object repeatedly [7]. Some studies have explored using

finger movements to rotate an object [11, 12], but they proposed a novel controller rather than utilizing a conventional VR controller.

We considered using a trackpad in a VR controller to improve the current 3D manipulation techniques. The trackpad is built-in to most commercial VR controllers. We could expect the users to perform better and lower their physical fatigue when rotating objects by reducing clutching, while still allowing them to employ the virtual hand technique that they are skilled with. As a result, we applied the virtual trackball method for 3D rotation with the trackpad. The virtual trackball method is a well known method for implementing 3D rotation with 2D input devices [4, 5].

In this study, we implemented two representative virtual trackball methods—Arcball and Two-axis Valuator—with a commercial VR controller and conducted a user study to compare their performance and usability of them. The contribution of this work is twofold: 1) The first attempt to apply virtual trackball methods on a commercial VR controller form factor, 2) Empirical findings on the performance and usability of the suggested methods.

2 RELATED WORK

2.1 3D Manipulation Techniques for Rotation in Virtual Reality Environments

Some non-isomorphic rotation techniques have been proposed in various virtual environments to enhance 3D rotation efficiency [13, 16]. Poupyrev et al. [13] demonstrated the effectiveness of non-isomorphic rotational mappings in the 3D interface. They showed the effectiveness of a static scale of 1.8 with a ball-shaped 3D input device. Dynamic non-isomorphic methods also have been suggested to improve rotation performance. Frees et al. [6] proposed an interaction technique called PRISM, which adjusts the control/display (C/D) gain according to the user's current hand speed. Gao et al. [7] compared the effects of static and dynamic mapping under the same conditions using a recent VR system and found that only dynamic non-isomorphic mapping significantly improved rotation performance compared to isomorphic mapping.

Non-isomorphic mapping has been shown to be efficient for rotation, but there are some limitations. Non-isomorphic rotation differs from actual rotation, making it unnatural. Furthermore, the availability or effect on distant objects has yet to be explored. It was only proven with direct manipulation [7].

2.2 Virtual Trackball Techniques for 3D Rotation

Numerous 3D rotation techniques using a 2D control device have been proposed, but virtual trackball methods are the most commonly used techniques in commercial applications [4]. The virtual

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trackball allows users to manipulate 3D objects like rolling a sphere directly. Chen et al. [5] proposed the Virtual Sphere as the first virtual trackball technique to emulate the mechanics of a physical 3D sphere. The user's 2D motion is mapped to the 3D virtual sphere behind the 2D plane, allowing the user to interact as if rotating the 3D sphere. However, the rotation axis is not necessarily perpendicular to the 2D motion vector, leading to unnatural rotation. Bell [2] proposed an improved virtual trackball technique that maps 2D motion to a combination of a sphere and a hyperbola, resulting in smooth rotation. Shoemake [15] proposed a new virtual trackball called Arcball. It provides a kinetic agreement of 2D motion and object rotation, allowing users to make more natural rotations. The Two-axis Valuator [1] (originally called continuous XY with Additional Z controller [5]) technique maps a vertical or horizontal 2D movement to a vertical or horizontal rotation of a 3D object [1]. Unlike the previous techniques, this does not utilize geometric property, and the same movement always produces the same rotation no matter where the movement starts. Despite various virtual trackball techniques proposed and compared by researchers, the results have been inconsistent and have not provided clear guidance. [14]. We implemented representative virtual trackball techniques and evaluated their performance to identify the most effective virtual trackball for use in a VR controller.

3 VIRTUAL TRACKBALL ON VR CONTROLLER

Previous literature has revealed that Bell's Virtual Trackball [2], Shoemake's Arcball [15], and the Two-axis Valuator [1] as the most representative methods [14]. In our pilot study, the performance difference between Bell's Virtual Trackball and Shoemake's Arcball was found to be negligible. We selected Shoemake's Arcball as a candidate because it enables more intuitive movement than Bell's Virtual Trackball. Thus, we decided to implement Shoemake's Arcball and the Two-axis Valuator on the VR controller and compare these with a regular VR controller. However, due to the higher intuitiveness of Shoemake's Arcball, we chose it as a candidate for our study. We subsequently implemented both Shoemake's Arcball and the Two-axis Valuator on the VR controller and compared their performance against that of a standard VR controller.

3.1 Virtual Trackball

3.1.1 Shoemake's Arcball. Fig. 1 shows the model of the Arcball. Point p on the image plane is mapped to point P of the 3D sphere surface. The quaternion for rotation is determined by P_a and P_c mapped from the touch-down point p_a and the touch-up (or touch) point p_c . The axis of rotation is perpendicular to the plane containing the vector $\overrightarrow{OP_a}$ and the vector $\overrightarrow{OP_c}$, and the angle of rotation is twice the angle between them.

We designed a model in which a trackpad is placed on an image plane, and a 3D virtual sphere with the same radius is positioned behind the image plane. As a result, all points on the trackpad are projected onto the inside of the sphere. The radius of the sphere was set to 1. Consequently, the mapping function $f_{Arcball}$ of point P on the surface of the sphere from point $p = (x, y, 0)$ on the trackpad is

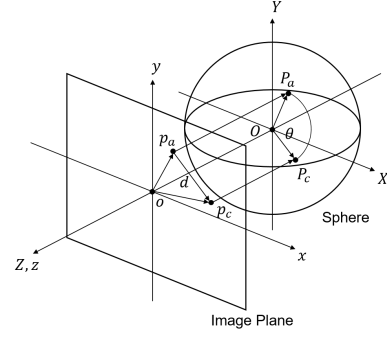
$$f_{Arcball}(x, y, 0) = P = (x, y, \sqrt{1 - (x^2 + y^2)}), (x^2 + y^2 \text{ is always less than or equal to } 1)$$


Figure 1: A virtual trackball can be conceptualized as a sphere located behind the screen. Image reproduced from [9].

3.1.2 Two-axis Valuator. The Two-axis Valuator technique maps a vertical and horizontal movement on the trackpad to the rotation about X-axis (horizontal axis) and Y-axis (vertical axis). Diagonal movement is mapped to the rotation about an axis that is a combination of the two axes so that the axis of rotation and motion are perpendicular. Hence its movements are highly predictable. Furthermore, unlike Arcball, the rotation does not change depending on the starting point. In other words, the same displacement always produces the same rotation. The end-to-end movement of the trackpad was designed to rotate 180 degrees, similar to Arcball.

Although the Two-axis Valuator technique provides predictable movements, it is not able to rotate about the Z-axis (forward axis). Previous studies utilized the outside of the circle area [14] or the mouse wheel [18] for the Z-axis rotation. However, the effectiveness of these approaches was not clear, and they are difficult to apply to a VR controller. Thus, we decided to use a simple Two-axis Valuator.

3.2 Implementation on VR controller

We devised a method for manipulating virtual objects using a virtual trackball on a trackpad. A user can rotate the object by swiping the trackpad with their thumb while holding the object using the trigger button with their index finger. We translated the trackpad plane so that the midpoint of the trackpad plane coincides with the midpoint of the object, giving the user the impression of controlling the trackball directly. To aid users in understanding the operation of each method, we implemented visual feedback (Fig. 2). For the Arcball, we provided visual feedback showing the point projected onto the 3D sphere from the touchpoint on the trackpad. For the Two-axis Valuator, two axes of rotation are shown.

4 USER STUDY

The study aims to compare the performance of three rotation techniques (Regular VR controller, VR controller using the Arcball, and VR controller using the Two-axis Valuator). Bergström et al. [3] recommend deciding early on whether the goal of the study is to evaluate speed or accuracy. Thus, we defined our main goal as evaluating speed performance and designed the study accordingly.

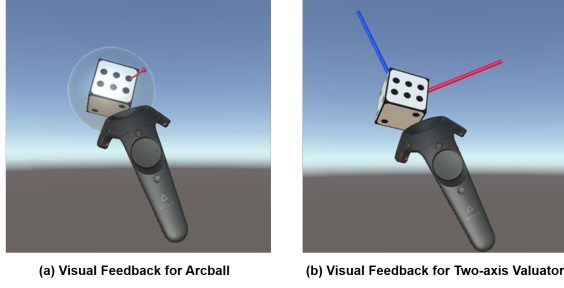


Figure 2: The visual feedback of each method. (a) Arcball provides visual feedback showing the projected point from the touchpoint onto the 3D sphere. (b) Two-axis Valuator provides two axes of rotation.

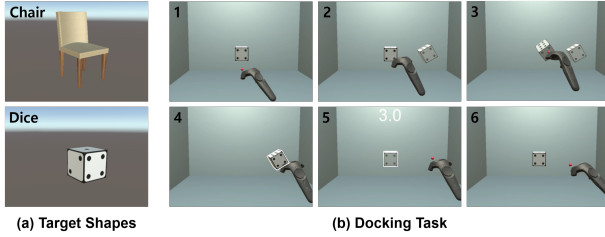


Figure 3: (a) Target shapes used in our user study. (b) The process of a docking task in our user study.

4.1 Participants

We recruited 18 participants ($M = 21.11$, $SD = 2.08$, 7 female and 11 male) from the university's online community. All participants were right-handed. Twelve participants answered that they had experienced VR content before. Among them, two participants answered that they used VR devices for an hour every week, and 10 answered that they had only a few experiences.

4.2 Experimental Setup

We chose HTC VIVE Pro Eye for the VR HMD. Participants sat on a chair, wore the VR HMD, and held the VIVE controller in their right hand. The HMD and controller were tracked by two VIVE base stations installed about 1.5m in front. The area that participants' arms could reach was empty.

4.3 Task

We designed a six-degrees of freedom (6-DOF) docking task to compare the performance of rotation techniques. Two target shapes were chosen due to the possibility of task performance results being influenced by the required degree of rotation. We chose a chair and a dice as docking objects (Fig. 3(a)). The chair has low ambiguity of orientation, so a user can easily identify the target rotation [17]. On the other hand, the user needs to scan each side of the dice to complete the docking task since it has a symmetrical shape. The size of the chair and the dice are about 10 cm × 10 cm × 20 cm and 7 cm × 7 cm × 7 cm, respectively.

The docking task was conducted in the following steps (Fig. 3(b)). A chair (or a dice) appears at the center when the task starts. Users can grab the chair by pressing the trigger with their index finger. A translucent target appears as soon as the user grabs the chair. The user matches the chair to the target until a white outline is displayed. The outline appears when the chair aligns with the target within a certain error threshold. The error thresholds for position and orientation were 1.5 cm and 8 degrees, respectively. After one trial, a chair appears in the center again in 3 seconds.

We defined the target's position and orientation in advance to ensure the same target set for each condition. The targets are rotated from the original object's orientation around the following axis vectors: x (1,0,0), y (0,1,0), z (0,0,1), x-y (1,1,0), x-z (1,0,1), and y-z (0,1,1) axis. They are rotated around the axes 150 degrees, either clockwise or counterclockwise. The docking target appears on the left or right side, 20 cm from the center. Therefore, a total of 24 targets (6 axes × 2 directions × 2 positions) were pre-determined. The targets appeared in random order.

4.4 Design & Procedure

We used a within-subject design with *Rotation Technique* (VR Controller, Arcball, Two-axis Valuator) as an independent variable. The order of *Rotation Technique* was fully counterbalanced among the participants. For each *Rotation Technique*, the participants were given a brief explanation and had 5 minutes of practice. The Arcball and Two-axis Valuator showed visual feedback in the first half of the practice session. Participants performed 24 docking tasks for each shape after practice. All participants performed the task on a chair shape followed by a dice shape. There were one-minute breaks between practice and each shape task. After one *Rotation Technique* condition was completed, raw NASA-TLX [8] was collected for task load measurement and a three-minute break was given. We collected data from a total of 2,592 trials (18 participants × 3 techniques × 2 shapes × 24 tasks). After the experiment, we collected the preference rank of each *Rotation Technique* and conducted a brief interview. We measured task completion time, object rotation angle, and the number of clutching.

5 RESULTS

We conducted a one-way repeated measures ANOVA test with a Greenhouse-Geisser adjustment (depending on sphericity assumption) and a post hoc t-test with Bonferroni correction for a parametric test. Since only the completion time on the dice shape satisfies the ANOVA assumption, the parametric test was conducted only on the completion time on the dice shape. We conducted a Friedman test and a post hoc Wilcoxon Signed Rank test with Bonferroni correction for a non-parametric test. Corrected p-values were reported for a post hoc test.

5.1 Completion Time

Fig. 4(a) shows the average task completion time for each *Rotation Technique*. For the chair task, a Friedman test on completion time showed a significant main effect of *Rotation Technique* ($\chi^2(2) = 7.000$, $p < 0.001$). Post hoc test results showed that the Arcball was significantly faster than the VR Controller ($p = 0.017$). No significant difference was found in the other pairs. For the dice task,

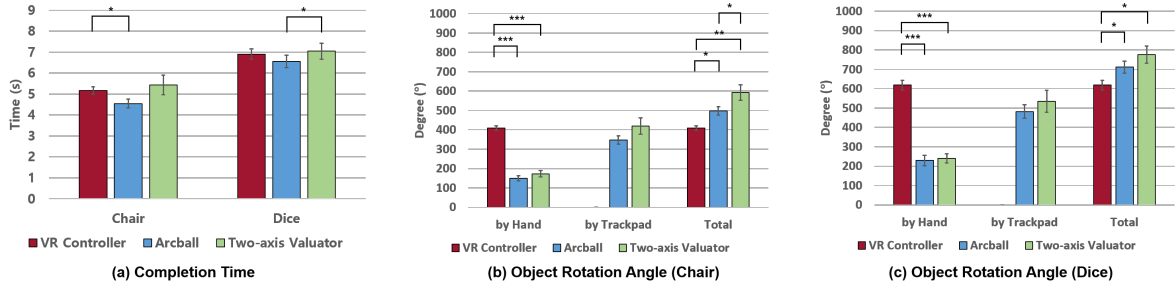


Figure 4: Results in our user study. From left to right: (a) completion time, (b) object rotation angle for the chair target, and (c) object rotation angle for the dice target. Error bars indicate standard errors, and asterisks indicate significant differences (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$).**

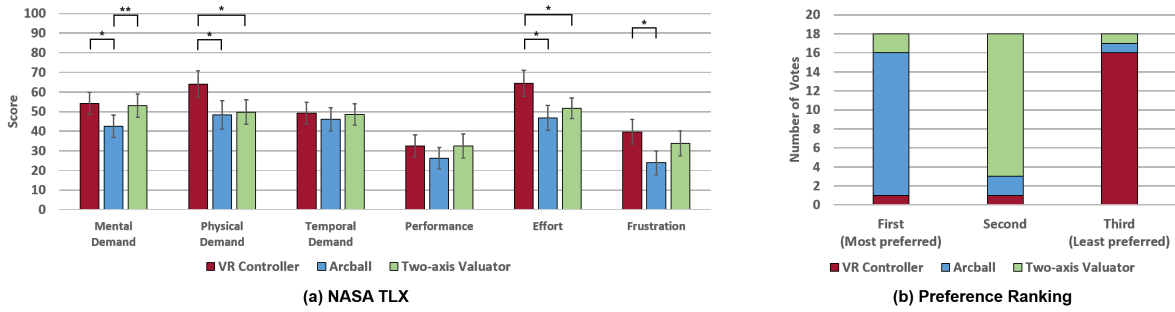


Figure 5: Results in our user study; (a) NASA TLX score and (b) Preference survey results. Error bars indicate standard errors, and asterisks indicate significant differences (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$).**

an ANOVA analysis showed a significant main effect of *Rotation Technique* ($F(2, 34) = 4.528$, $p = 0.009$, $\eta^2 = 0.436$). Post hoc test results showed that the Arcball was significantly faster than the Two-axis Valuator ($p = 0.028$). No significant difference was found in the other pairs.

5.2 Object Rotation Angle

Fig. 4(b-c) shows the average rotation angles for each *Rotation Technique*. The object rotation angle indicates the amount of object rotation during a task. We separated the object rotation angle into the rotation performed by the arm and wrist movement (by hand) and by the thumb movement (by trackpad), and analyzed them respectively. For the chair task, a Friedman test showed that the main effect due to *Rotation Technique* significantly affected the object rotation angle ($\chi^2(2) = 16.000$, $p < 0.001$) and object rotation angle by hand ($\chi^2(2) = 24.778$, $p < 0.001$). For the dice task, a Friedman test showed that the main effect due to *Rotation Technique* significantly affected the object rotation angle ($\chi^2(2) = 8.444$, $p = 0.015$) and object rotation angle by hand ($\chi^2(2) = 27.444$, $p < 0.001$). Post hoc results are shown in the Fig. 4(b-c).

5.3 Clutching Count

For the chair task, the means of clutching counts were 3.535 for the VR Controller, 1.544 for the Arcball, and 1.775 for the Two-axis Valuator. A Friedman test showed a significant main effect of *Rotation Technique* ($\chi^2(2) = 25.042$, $p < 0.001$). Post-hoc results found that

the VR Controller had significantly more clutching counts than the Arcball ($p < 0.001$) and the Two-axis Valuator ($p = 0.002$). For the dice task, the means of clutching counts were 4.234 for the VR Controller, 1.991 for the Arcball, and 2.292 for the Two-axis Valuator. A Friedman test showed a significant main effect of *Rotation Technique* ($\chi^2(2) = 20.817$, $p < 0.001$). The VR Controller had significantly more clutching counts than the Arcball ($p < 0.001$) and the Two-axis Valuator ($p < 0.001$).

5.4 Task Load & Preference

Fig. 5(a) shows the average NASA TLX score for each *Rotation Technique*. A Friedman test showed that the rotation technique significantly affected Mental Demand ($\chi^2(2) = 12.746$, $p = 0.002$), Physical Demand ($\chi^2(2) = 0.656$, $p = 0.013$), Effort ($\chi^2(2) = 12.230$, $p = 0.002$), and Frustration ($\chi^2(2) = 12.133$, $p = 0.002$). No significant differences were found for Temporal Demand ($\chi^2(2) = 1.019$, $p = 0.601$) and Performance ($\chi^2(2) = 3.633$, $p = 0.163$). Post hoc results are shown in the Fig. 5(a). Fig. 5(b) shows preference rank results for each *Rotation Technique*. Most participants preferred the Arcball, the Two-axis Valuator, and the VR controller in that order. A Friedman test showed that there was a statistically significant difference ($\chi^2(2) = 23.444$, $p < 0.001$). The post hoc result showed that all pairs are significant.



Figure 6: A distant object manipulation by combining the virtual trackball and the raycasting techniques. (1-2) Users can select and translate a distant chair using the raycasting. (3) After translating the chair, the users can adjust the angle of the chair using the virtual trackball.

6 DISCUSSION

Our results show that Arcball is generally better than a regular VR controller. However, the Two-axis valuator was not effective. We discuss specific findings in the following sub-sections.

6.1 Effect of the Virtual Trackball on a VR Controller

The participants could significantly reduce the arm and wrist movement and clutching counts using the Arcball and the Two-axis Valuator. It was also reflected in the physical demand score. However, the object rotation angle increased when using the virtual trackball. Direct manipulation using a VR controller provided a totally predictable rotation, but the participants were unfamiliar with the virtual trackball, so they appeared to perform an inefficient rotation.

The results of performance varied depending on the rotation technique and target shape. The Participants completed tasks faster with the Arcball than with the VR controller for the chair task but not for the dice task. We predicted that the virtual trackball would be more effective for the dice task requiring more rotation, but it was only effective for the chair task where the target rotation was easily identified. The participants had to scan all sides for the dice task. It seemed that the scanning process was slower due to their lack of familiarity with the virtual trackball.

Most participants preferred the virtual trackball methods to the VR controller, but one participant (P8) preferred the VR controller the most. The participant responded that the rotation of the virtual trackball was difficult to understand. This was also reflected in the performance. This participant took an average of 11.7 seconds for the chair task and 10.5 seconds for the dice task using the Two-axis Valuator.

6.2 Comparison of the Arcball and the Two-axis Valuator

Our results indicated that the Arcball outperformed the Two-axis Valuator, as reflected in the preference rankings. Most participants reported responded they could easily achieve the desired rotation using the Arcball, while there were cases where it was challenging to achieve the intended rotation when using the Two-axis Valuator. In the NASA TLX results, the Two-axis Valuator showed significantly higher mental demand and effort than the Arcball. However, none of the participants could elaborate on the difference between the two conditions. We speculated that the ability to rotate around

the Z-axis by turning the edge of the trackpad using the Arcball appeared to be more naturally acceptable to the participants.

6.3 Application of the Virtual Trackball on a VR Controller

The virtual trackball's most distinct characteristic is its capability to rotate objects independently of the VR controller's 6-DOF input. This unique ability enables it to be utilized with the existing 6-DOF input technique. One possible application of the virtual trackball is its combination with raycasting for the translation and rotation of a remote object (Fig. 6). By utilizing the raycasting technique, users can select a remote object, while also having the ability to adjust the ray's length by pressing the trackpad with their thumb and using it as a D-pad. After translating the object, the users can adjust the angle of the object by swiping the trackpad with their thumb. The virtual trackball allows users to rotate a remote object without interfering with the raycasting.

6.4 Limitations & Future Work

In this study, participants used each technique for 10 to 15 minutes, including practice time. However, this duration may not be enough for them to become entirely proficient with the techniques. We noticed that the participants struggled to use the virtual trackball effectively and tended to over-rotate the object while performing the task. Future studies will need to investigate the effects of the virtual trackball when users get proficient with the virtual trackball.

Our results found that the Two-axis Valuator was not effective. Despite its potential for improvement, this study did not examine it further. One possible improvement for the Two-axis Valuator is an adjustable C/D gain, which is impossible with the Arcball. The C/D gain used in this study may not be optimal. Additionally, some 3D modeling tools such as Blender and Autodesk 3ds Max utilize a special version of the Two-axis Valuator called Two-axis Valuator with a fixed up-vector. This version uses the world's up-vector for horizontal rotation. The effect of these variations should also be explored.

7 CONCLUSION

In this study, we explored the effect of two virtual trackball methods on a VR controller. Our findings suggest that using the Arcball with a VR controller is more effective than a regular VR controller for rotating tasks. However, the Two-axis valuator was found to be ineffective. Both virtual trackball techniques significantly decreased

the wrist and arm movement required for the task. Participants preferred the Arcball over the other method. We believe that utilizing a virtual trackball with a VR controller has the potential for various applications, including distant object manipulation and rotational tasks.

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